

OOAD-LAB3-Activity Diagram

Name: M S SAMARTH BHARADWAJ

SRN: PES1UG23CS335

Section: F

Activity Diagram – Autonomous Drone Emergency Delivery System

Understanding of the Scenario:

This activity diagram represents the workflow of an Autonomous Drone Emergency Delivery System used during disaster situations. The process begins when a requester generates an emergency request. The Central Control System verifies the authenticity of the request and checks drone availability and weather conditions before assigning a delivery mission. The autonomous drone executes the delivery while monitoring its battery status until the mission is either completed or aborted.

Initial & Final Nodes:

The diagram starts with an initial node indicating the generation of an emergency request. The process ends at final nodes representing invalid request termination, waiting queue placement, mission abortion due to low battery, or successful delivery completion.

Action States:

The diagram includes action states such as request generation, verification, checking conditions, mission creation, flight path planning, payload preparation, system diagnostics, delivery execution, battery monitoring, mission update, and requester notification.

Decision & Merge Nodes:

Decision nodes evaluate request validity, drone and weather availability, and battery threshold during flight. Appropriate guard conditions ensure correct control flow and merging of activities.

Fork & Join (Parallelism):

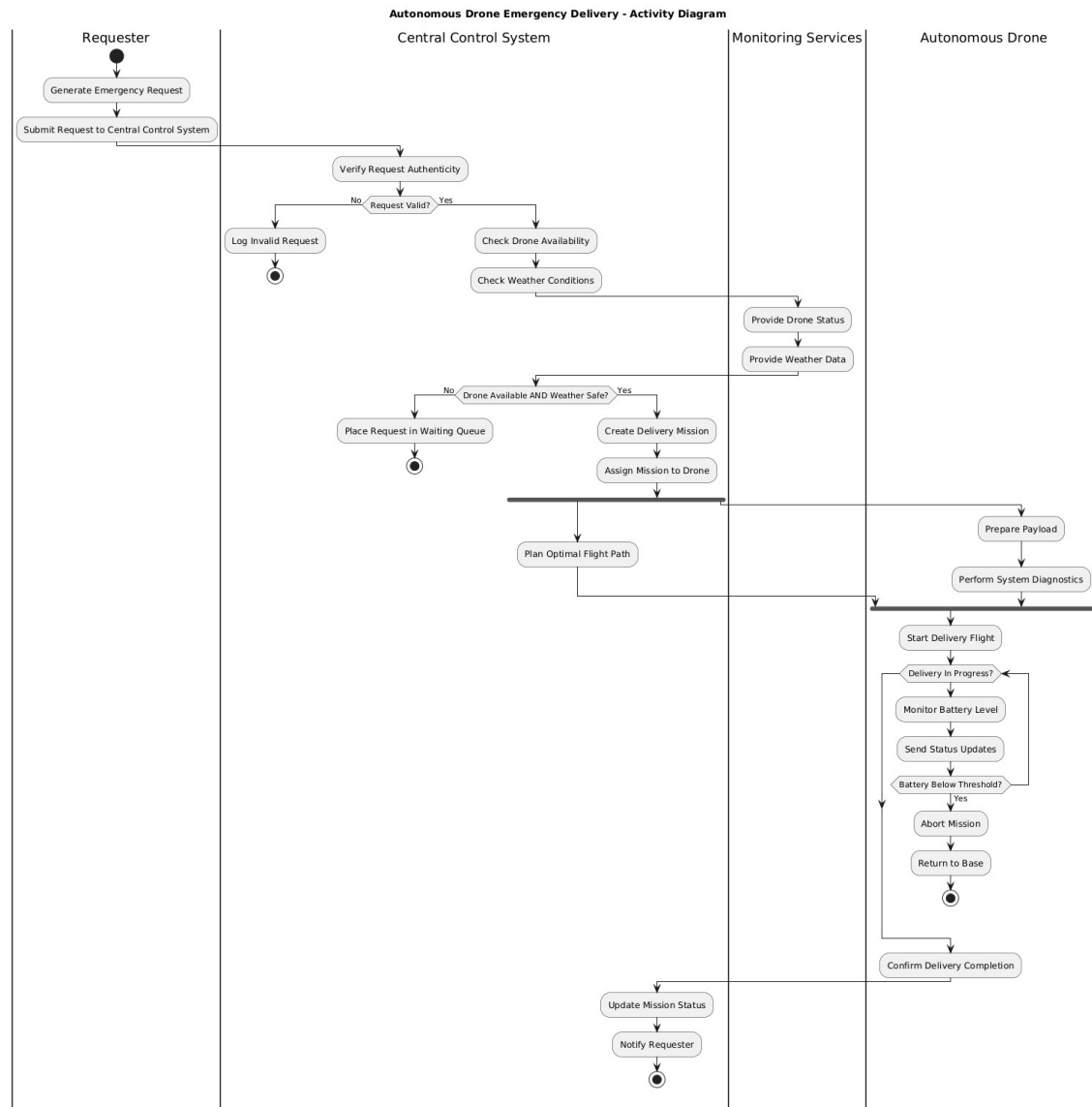
Parallel activities are represented using fork and join nodes where the Central Control System plans the optimal flight path while the Autonomous Drone prepares the payload and performs system diagnostics.

Swimlanes / Responsibility Separation:

Swimlanes clearly separate responsibilities among the Requester, Central Control System, Monitoring Services, and Autonomous Drone.

UML Notation & Diagram Clarity:

Standard UML activity diagram notations are used correctly, ensuring clarity, proper flow, and accurate representation of the system workflow.



Lab 5 – State Diagram

Online Railway Ticket Reservation System

Understanding of the Scenario:

This state diagram represents an Online Railway Ticket Reservation System. The system allows passengers to search for trains, select seats, and complete payment within a limited time. Once a seat is selected, it is temporarily blocked while internal activities and payment processing occur concurrently. Based on the outcome, the ticket is either confirmed or the seat is released.

Initial and Final States:

The system starts in the Idle state. Final states occur when the ticket is successfully issued or when the seat is released due to payment failure, timer expiration, or cancellation after refund processing.

Major States Description:

Idle – System waits for passenger interaction.

Searching – Passenger searches for available trains.

SeatSelected – Passenger selects a seat.

SeatBlocked – Seat is temporarily blocked and timer starts.

Confirmed – Payment completed successfully.

TicketIssued – Ticket is generated and issued.

Cancelled – Passenger cancels the confirmed ticket.

RefundProcessing – Refund is processed.

Released – Seat is released back to availability.

Composite and Concurrent States:

The SeatBlocked state is a composite state containing internal activities such as fare calculation and seat availability monitoring. Passenger validation and payment processing occur concurrently within this state.

Transitions and Conditions:

If payment succeeds within the allowed time, the ticket is confirmed and issued. If payment fails or the timer expires, the seat is released automatically. After confirmation, cancellation triggers refund processing and seat release.

UML Notation and Diagram Clarity:

Standard UML state diagram notations including initial and final states, composite states, concurrent regions, transitions, and internal activities are used correctly to ensure clarity and correctness.

Online Railway Ticket Reservation - State Diagram

